

NATCHAI SIRIMA

Environmental Manager | Environmental Engineer

Environmental Engineer with 12+ years of experience in railway, infrastructure, hydropower and renewable energy projects across Thailand, Laos, Turkey and the Philippines.



PERSONAL INFORMATION

Mobile	+66 87 656 2223 (TH)	Work / Philippines Mobile	+63 945 332 1831 (PH)
Email	natchai.itd@gmail.com	Nationality	Thai
Current Location	Pampanga, Philippines / Bangkok, Thailand	Website / Portfolio	https://natchaisirima.github.io/

PROFESSIONAL SUMMARY

Environmental Engineer with more than 12 years of experience on major infrastructure projects with a combined project value exceeding USD 1.6 billion, covering four countries and five major projects. Experienced in environmental management for internationally funded and standards-driven railway projects, including CEMMP implementation, EIA/EIS compliance, permit and regulatory coordination, environmental monitoring, waste, wastewater and hazardous waste management, and coordination with government agencies and project consultants. Also applies AI-assisted development to build and deploy the project's environmental data, reporting, monitoring and document-tracking tools.

EDUCATION

Master's Degree	N/A	Field / Major	-
Institution	-	Year Graduated	-
Bachelor's Degree	Bachelor of Engineering (B.Eng.)	Field / Major	Environmental Engineering
Institution	Chiang Mai University, Thailand	Year Graduated	2014

PROFESSIONAL ENGINEERING LICENSE

Level	Associate Engineer	Discipline	Environmental Engineering
License No.	นธ.3808	Council Member No.	240621
Date of Issue	29 Sep 2024	Date of Expiry	28 Sep 2029

COMPUTER & PROFESSIONAL SKILLS

Microsoft Word, Excel and PowerPoint - technical reporting, data tracking and project documentation
AutoCAD - drawing review and support for environmental engineering works
Adobe Photoshop - preparation and editing of report graphics and supporting documentation

AI-Assisted Development (Vibe Coding) - building web applications, automation and reporting tools with AI coding agents; Git/GitHub, cloud deployment and automated testing

Environmental Monitoring Instruments - noise and vibration, water/wastewater and field monitoring

CEMMP / EIA-EIS compliance, permit tracking, environmental reporting and regulatory coordination

Languages: Thai (Native), English (Professional Working Proficiency)

AI-ASSISTED DEVELOPMENT & DIGITAL TRANSFORMATION

- Applies AI-assisted development - commonly referred to as Vibe Coding - to design, build, test and deploy practical web applications, automation tools, monitoring systems, reporting workflows and management dashboards for environmental and construction operations. Combines environmental engineering domain expertise with modern AI coding agents to convert operational requirements into tested, usable digital solutions. Every system is specified against existing project procedures and official document formats, and is reviewed, tested and validated before staff rely on it.
- Developed and deployed a web-based environmental management portal covering permit, work inspection request, monitoring and activity registers, replacing multiple stand-alone spreadsheets with a single auditable record with renewal alarms.
 - Built and operate a site-specific weather forecasting and rainfall alert system for the project corridor, delivering scheduled forecast briefings and graded live rain alerts to site teams through group messaging and mobile push notifications.
 - Automated statutory and inspection reporting - including tree-cutting Terminal Reports with full annexes, Site Observation Reports and environmental inspection close-out responses - generating the required Word documents directly from structured data, photographs and Excel imports.
 - Delivered a document close-out tracking dashboard used by QA/QC and engineering staff, synchronising automatically from the source register while preserving manual corrections.
 - Developed mobile and desktop applications that stamp construction photographs with their nearest structure location, making photographic evidence filing-ready and usable offline in the field.
 - Applies AI/LLM capabilities where they add measurable value - drafting report text, extracting findings from inspection documents and summarising forecast data - always with deterministic fallbacks and human review before issue.
 - Uses AI coding agents for specification, implementation, debugging, automated testing and iterative improvement; work is version-controlled in Git and deployed to managed cloud services.

PROFESSIONAL EXPERIENCE - ITALIAN-THAI DEVELOPMENT PUBLIC COMPANY LIMITED (ITD)

North-South Commuter Railway Extension (NSCR-EX), Contract Package N-03

Job Site / Project Code	CP N-03 / NSCR-EX	Position	Environmental Manager (previously Environmental Engineer / Acting Environmental Manager)
Period	Nov 2020 - Present	Project Location	Angeles / Clark, Pampanga, Philippines
Project Value	USD 415 Million	Employer / Client	Department of Transportation (DOTr), Philippines

Project Description: Approximately 12 km of elevated railway including Angeles and Clark Stations. Responsible for contractor-side environmental management across the civil works alignment and major construction areas.

Key Responsibilities

- Lead implementation of the Contractor Environmental Management and Monitoring Plan (CEMMP) and ensure compliance with EIA/EIS commitments, ECC conditions and project requirements.
- Established and managed the Environmental Management Unit (EMU) from mobilization, including assignment of PCO/environmental staff and follow-up of corrective actions.
- Coordinate with DOTr, DENR-EMB, LGUs and GCR Consortium for site inspections, meetings, environmental submissions and closure of environmental observations.
- Manage and track environmental permits and regulatory requirements, including ECC, Permit to Operate, Discharge Permit and Hazardous Waste Generator ID.
- Manage DENR-accredited environmental monitoring covering noise and vibration, water/wastewater, air quality, solid and hazardous waste, and environmental compliance reporting.

- Manage community/environmental grievances (GRM) and environmental interfaces with construction activities to ensure timely corrective action and closure.
- Digitised the unit's environmental records - permits, work inspection requests, monitoring data and reporting - into a shared web portal with renewal alarms and a full audit trail.
- Deployed a corridor weather forecasting and rain-alert system together with document and reporting automation tools, now used in daily site operations by the environmental and QA/QC teams.

High-Speed Railway & Expressway Projects - Bid Preparation

Job Site / Project Code	Bangkok Office / Bid Phase	Position	Environmental Engineer
Period	Dec 2019 - Nov 2020	Project Location	Bangkok, Thailand (supporting Thailand & Myanmar bids)
Project Value	USD 280M (HSR) / USD 464M (Yangon PPP)	Employer / Client	State Railway of Thailand (SRT) / Ministry of Construction, Myanmar

Project Description: Supported bid preparation and mobilization planning for the Bangkok-Nong Khai High-Speed Railway, Section I, and Yangon Elevated Expressway PPP from the Bangkok office.

Key Responsibilities

- Prepared environmental documentation and supporting information for mobilization of the Bangkok-Nakhon Ratchasima / Bangkok-Nong Khai High-Speed Railway, Section I.
- Organized and compiled bid submittal documents for the Yangon Elevated Expressway PPP Project.
- Coordinated technical information, documentation and requirements with bid/engineering teams to ensure complete and timely submissions.

Nam Theun 1 Hydropower Project

Job Site / Project Code	NT1 / CMC-ITD-SONG DA JV	Position	Environmental Engineer
Period	Dec 2016 - Dec 2019	Project Location	Pakkading, Bolikhamsai, Lao PDR
Project Value	USD 512 Million	Employer / Client	Nam Theun 1 Power Company Limited (NT1PC)

Project Description: 650 MW hydropower project featuring a Roller Compacted Concrete (RCC) curved gravity dam with a maximum height of approximately 177 m on the Nam Kading River.

Key Responsibilities

- Developed and managed the Environmental Management Plan for dam construction and supporting works.
- Conducted environmental monitoring covering noise and vibration, wastewater, surface water and waste management.
- Designed and supervised landfill/waste management areas, including site inspections and corrective action follow-up.
- Prepared environmental compliance reports and supported environmental audits and inspections for the project.

OTHER PROFESSIONAL EXPERIENCE

Bioclean International Co., Ltd. - Biogas Plant Projects

Job Site / Project Code	N/A	Position	Environmental Engineer
Period	Mar 2014 - Nov 2016	Project Location	Thailand & Turkey
Project Value	Approx. USD 5 Million	Employer / Client	Private / Renewable Energy Projects

Project Description: Design, operation and commissioning of biogas power plants using agricultural residues and animal manure in Thailand, with commissioning assignments in Turkey.

Key Responsibilities

- Designed and operated biogas plants in Ratchaburi and Kanchanaburi, Thailand.
- Worked in Turkey for approximately four months to commission fertilizer-processing equipment and support biogas plant installation at Çubuk and Efeler.
- Supported power generation processes using methane produced from agricultural waste and fresh manure.

TRAINING & PROFESSIONAL DEVELOPMENT

Verified training and professional development records based on the certificates provided.

Course / Training	Knowledge Sharing Session on Environmental Management Audit: Hazardous Materials and Wastes	Year	2026
Duration	Session held on 05 May 2026	Training Institution	GCR Consortium, in collaboration with Acciona-Daelim E&C Joint Venture
Course / Training	Environmental Training for Managing Heads	Year	2021
Duration	8 hours (10 May 2021) Certificate No. MH-068-2021	Training Institution	Dreamquest Consulting and Training Services

PROFESSIONAL ENGINEERING LICENSE



Professional Engineering License - Associate Engineer, Environmental Engineering | License No. ภส.3808 | Member No. 240621 | Valid until 28 Sep 2029